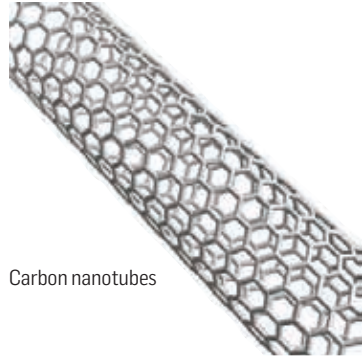


achieved by higher activity and greater selectivity in reactors and separation processes, to enabling small-scale distributed processing (making chemicals as close as possible to the point of use). There's already a move in the chemical industry towards this. Another use could be the small-scale, on-site production of high value chemicals such as pharmaceuticals.

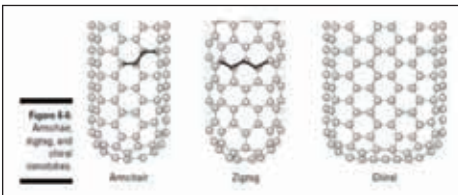
Two dimension Carbon Nanotubes

As far back as 1959, Roger Bacon had produced images of carbon nanotubes. In the 1980s, Howard Tennant applied for a patent for a method to produce them. In 1990, Richard Smalley postulated the concept that if buckyballs get big enough, they become carbon cylinders. But it wasn't until 1991 that Sumio Iijima, a researcher at NEC's Fundamental Research Lab, not only took photos of nanotubes, but also put two and two together to explain what nanotubes actually are — and put a name to them.



Carbon nanotubes

CNTs are nothing but extended tubes of rolled graphene sheets. Nanotubes come in a couple of varieties. They can either be single-walled carbon nanotubes (SWNT) or multiwalled carbon nanotubes (MWNT). As you might expect, an SWNT is just a single cylinder, whereas an MWNT consists of multiple concentric nanotube cylinders,



Different arrangements of CNT

One property of nanotubes is that they're really, really strong. Tensile strength is a measure of the amount of force an object can withstand without tearing apart.

The tensile strength of carbon nanotubes is approximately 100 times greater than that of steel of the same dia.

Nanotubes are strong but are also elastic. This means it takes a lot of force to bend a nanotube, but the little guy will spring right back to its original shape when you release it, just like a rubber band does. Young's modulus for carbon